

*Same as well
Just better*

SAME YOU SINCE 2009
JUST BETTER

Before we start

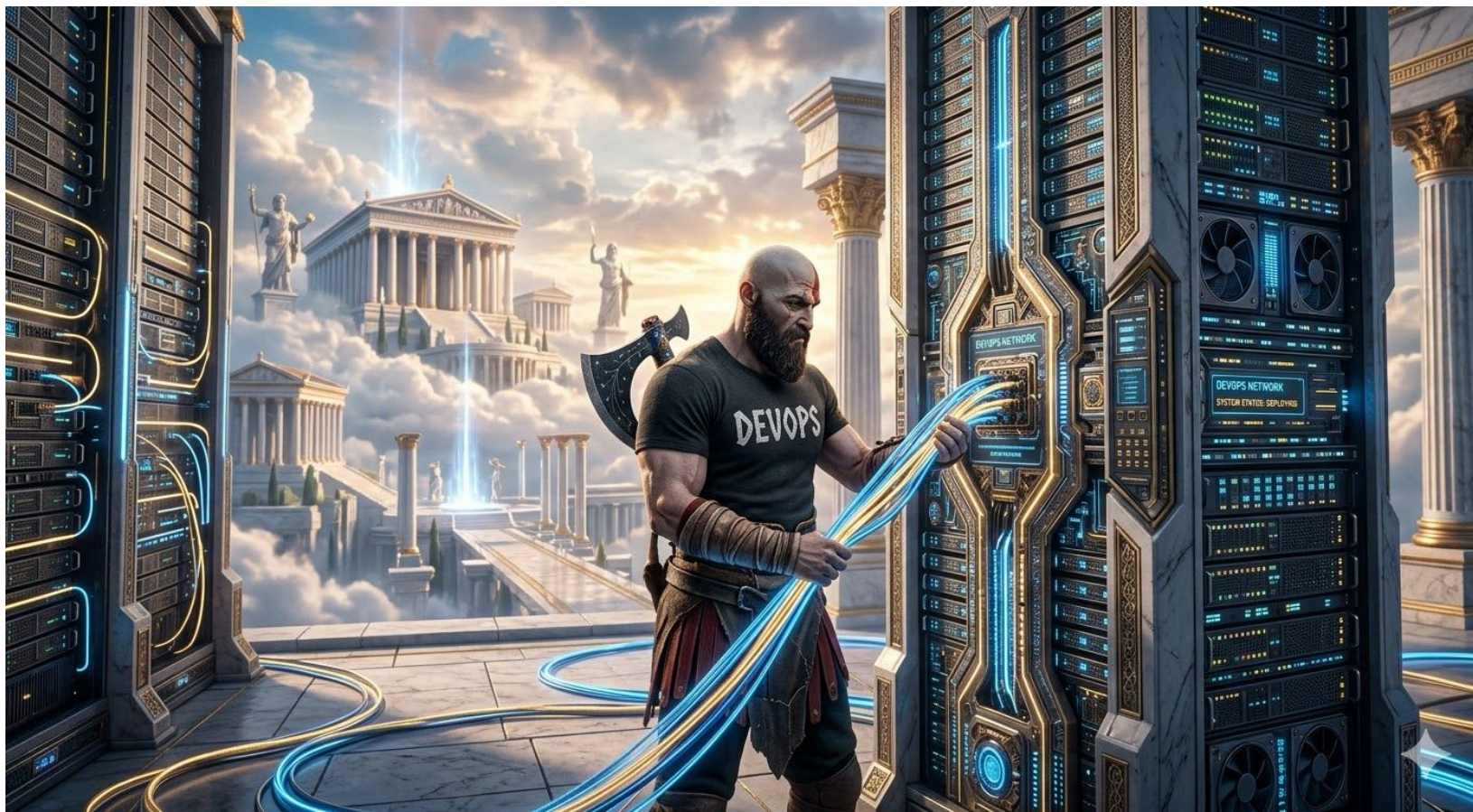
Project Genesis Check-In

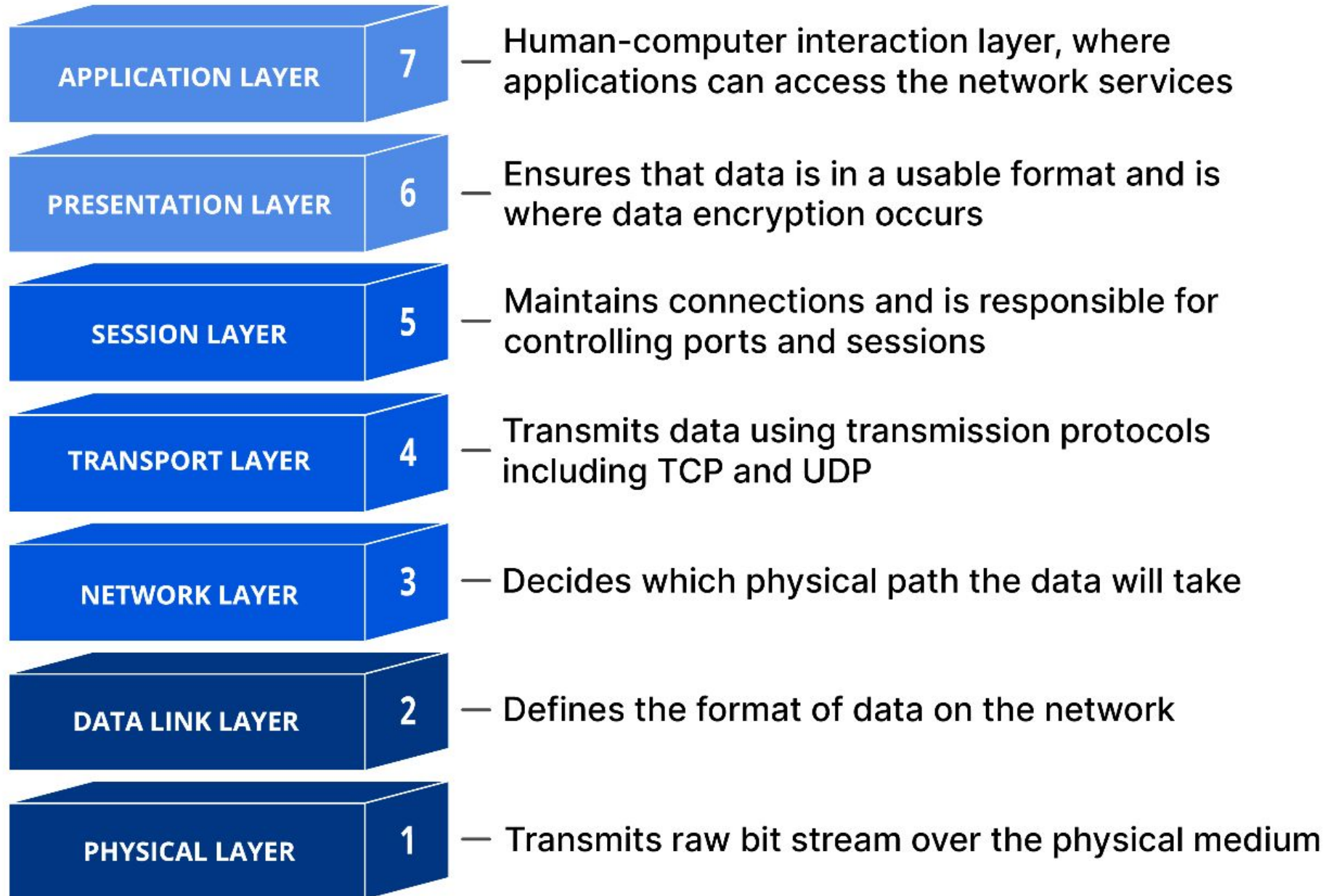
- You have a running web framework (FastAPI/Flask/Django).
- You successfully integrated a Database (SQL/NoSQL) using an ORM.
- You collaborated via branching and Pull Requests.
- The Problem: Your app runs locally, but how does the outside world actually reach it?

Lesson 8

Networking

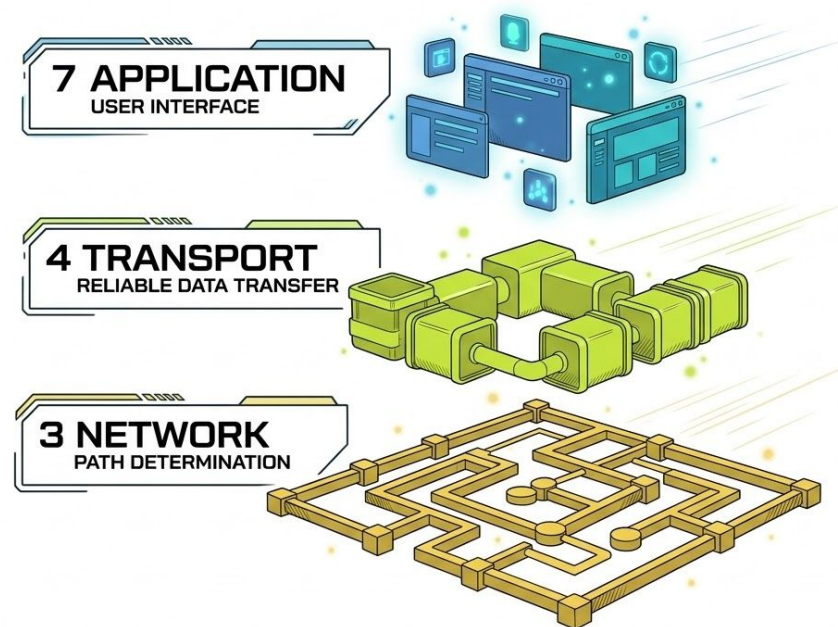
Networking is crucial for DevOps to ensure secure and reliable code accessibility. Without it, configuring Cloud VPCs, Load Balancers, or Kubernetes Ingress is impossible. We will explore the path from a user's browser to your application's health endpoint.





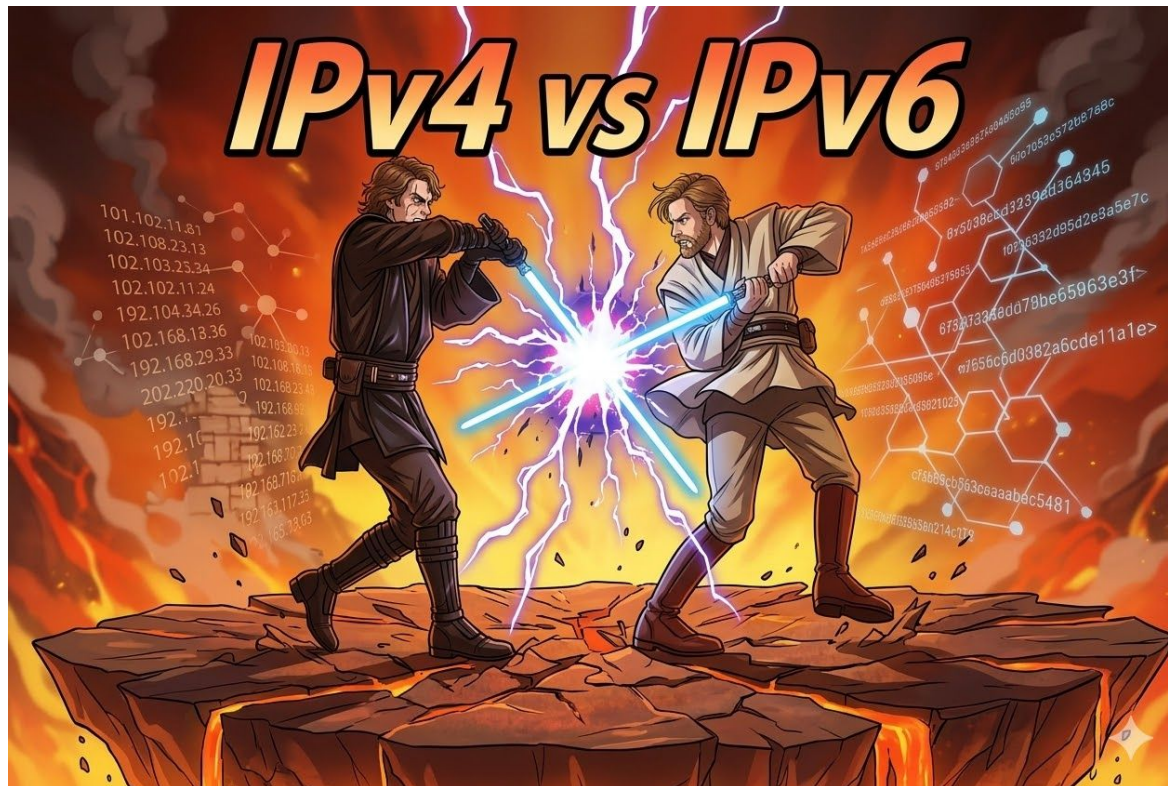
The OSI model is a conceptual framework describing how networks communicate.

- Layer 3 (Network): IP Addresses and Routing (Where is the server?).
- Layer 4 (Transport): TCP/UDP and Ports (Which application on the server?).
- Layer 7 (Application): HTTP/HTTPS, DNS (What is the actual payload/request?).



IPv4 vs. IPv6

- IPv4: 32-bit addresses (e.g., 192.168.1.1). The standard we will focus on. Limited to ~4.3 billion addresses.
- IPv6: 128-bit addresses. Created because we ran out of IPv4 addresses.
- In modern Cloud environments, you will configure both, but IPv4 remains the foundational knowledge for VPCs.



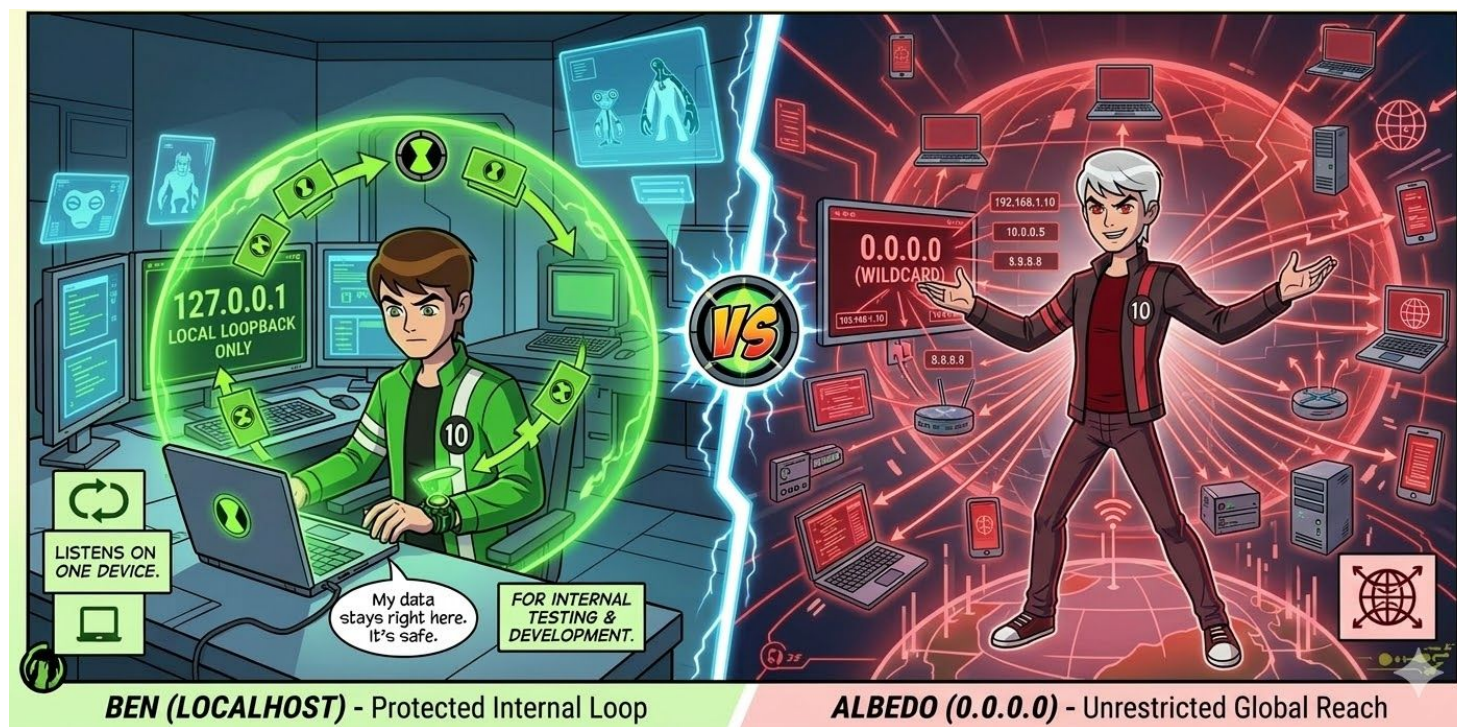
Public vs. Private IPs (RFC 1918)

- Public IPs: Routable on the open internet. Must be unique globally (e.g., 8.8.8.8).
- Private IPs: Used inside local networks (your home router, AWS VPCs). Not reachable from the outside without a gateway.
- Common Private Ranges: 10.x.x.x, 172.16.x.x to 172.31.x.x, 192.168.x.x.

First Address	Last Address	Common Mask
10.0.0.0	10.255.255.255	/8 (255.0.0.0)
172.16.0.0	172.31.255.255	/16 (255.255.0.0)
192.168.0.0	192.168.255.255	/24 (255.255.255.0)

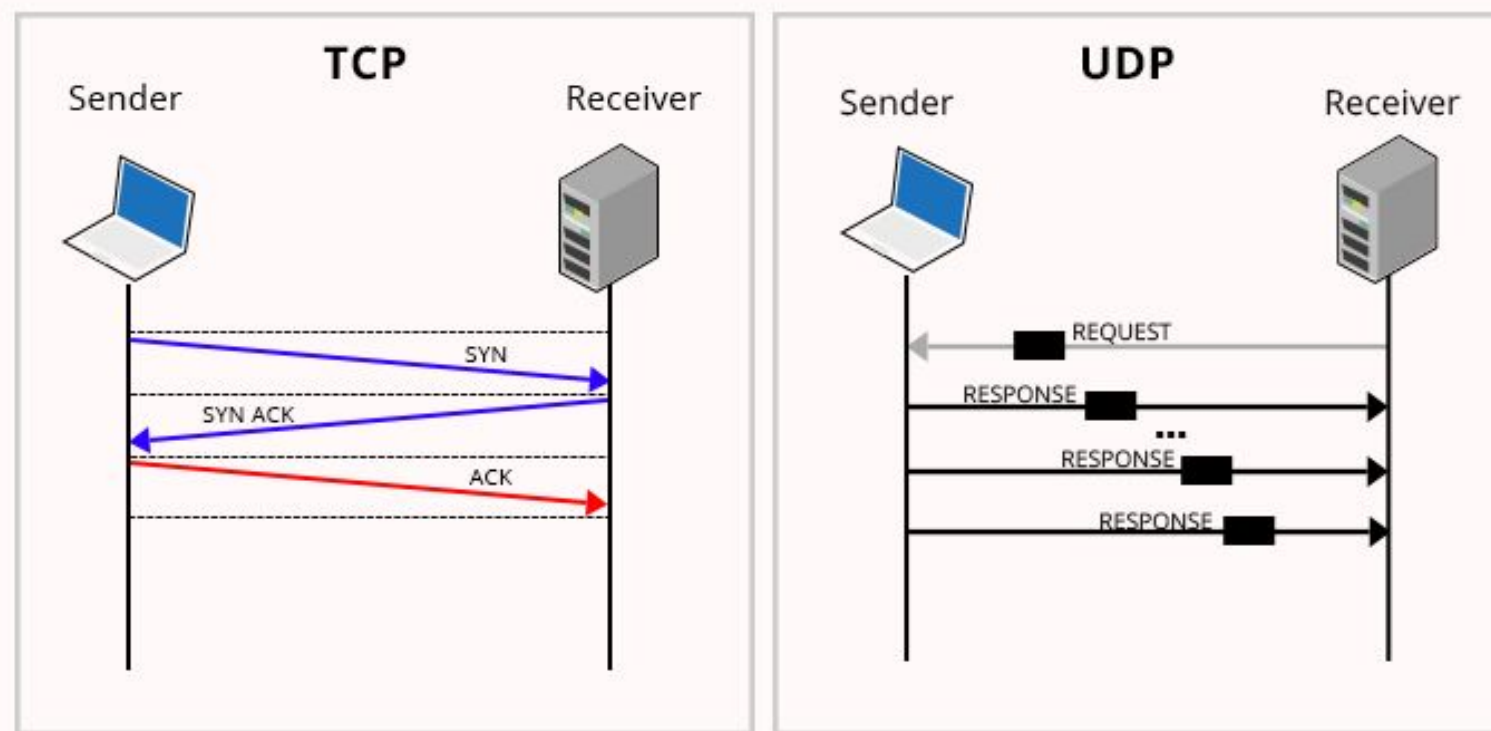
Localhost vs. 0.0.0.0

- 127.0.0.1 (Localhost): The loopback interface. The app only talks to itself. External requests are rejected.
- 0.0.0.0 (All Interfaces): Tells the server to listen for requests on every available network interface (including public IPs).
- Crucial for next week: If your Docker app listens on 127.0.0.1, you won't be able to reach it from outside the container!



Ports: The Application Doors

- If an IP address is a building's street address, the Port is the specific apartment number.
- Servers have 65,535 possible ports.
- Well-Known Ports: 80 (HTTP), 443 (HTTPS), 22 (SSH), 5432 (PostgreSQL), 3306 (MySQL).
- FastAPI defaults to 8000; Flask to 5000.



What is a Subnet?

- A Subnet (Subnetwork) is a logical subdivision of a larger IP network.
- Instead of having 60,000 servers in one massive chaotic network, we divide them into smaller, manageable, secure chunks.
- Example: One subnet for Public Web Servers, another hidden subnet for our Databases.

Class A (1-126)
Default subnet mask

<u>Network</u>	<u>Host</u>	<u>Host</u>	<u>Host</u>
255	0	0	0

Class B (128-191)
Default subnet mask

<u>Network</u>	<u>Network</u>	<u>Host</u>	<u>Host</u>
255	255	0	0

Class C (192-223)
Default subnet mask

<u>Network</u>	<u>Network</u>	<u>Network</u>	<u>Host</u>
255	255	255	0



- CIDR is the modern way to specify IP addresses and their routing prefix.
- Format: IP_Address/Prefix_Length (e.g., 192.168.1.0/24).
- The "Slash" indicates how many bits are locked for the network. The remaining bits dictate how many IP addresses are available for servers.

IPv4 CIDR IP/CIDR	to last IP address	Mask	Hosts (*)	Class
a.b.c.d/32	+0.0.0.0	255.255.255.255	1	1/256 C
a.b.c.d/31	+0.0.0.1	255.255.255.254	2	1/258 C
a.b.c.d/30	+0.0.0.3	255.255.255.252	4	1/64 C
a.b.c.d/29	+0.0.0.7	255.255.255.248	8	1/32 C
a.b.c.d/28	+0.0.0.15	255.255.255.240	16	1/16 C

- IPv4 has 32 bits total.
- Formula for available IPs: $2^{(32 - \text{Prefix})}$
- Example: A /24 subnet -> $2^{(32 - 24)}$ -> 2^8 -> 256 IPs total.
- Note: AWS always reserves 5 IPs per subnet, so a /24 gives you 251 usable IPs for your instances.

Common CIDR Cheat Sheet

CIDR 	Subnet Mask	Total IPs	Usable Hosts
/32	255.255.255.255	1	1 (Single Host)
/30	255.255.255.252	4	2
/28	255.255.255.240	16	14
/24	255.255.255.0	256	254
/20	255.255.240.0	4,096	4,094
/16	255.255.0.0	65,536	65,534

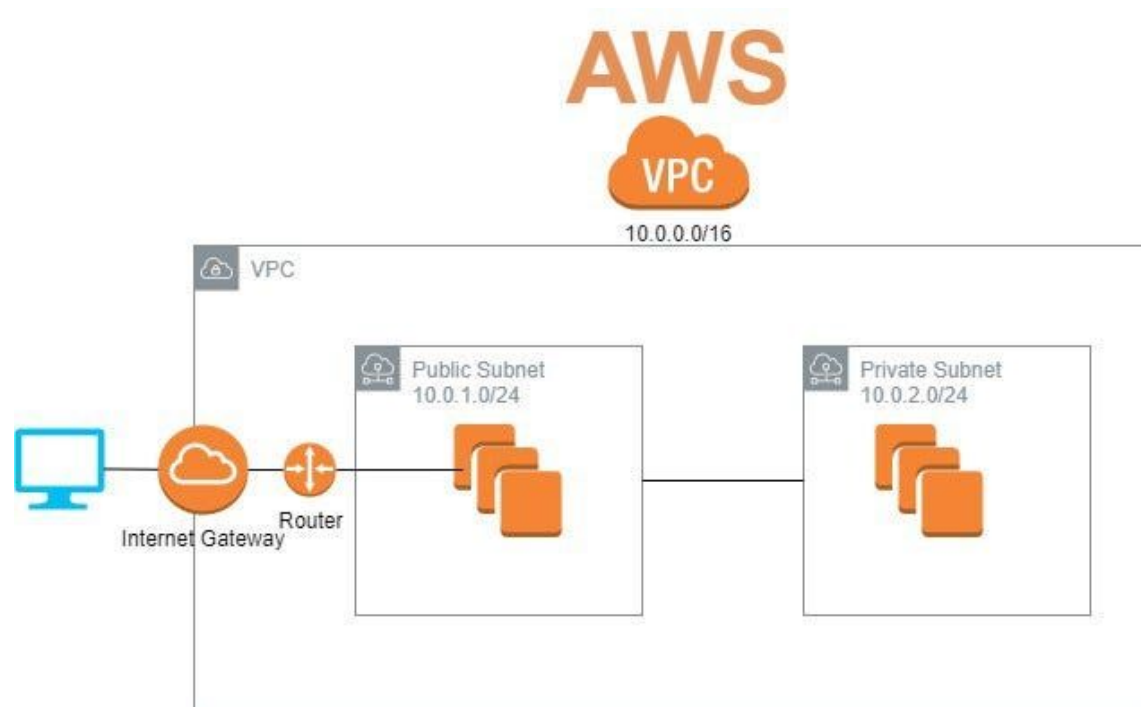
Core Calculations for IPv4

An IPv4 address consists of **32 bits**.

- **Total IP Addresses:** Calculate $2^{(32 - \text{prefix})}$.
 - Example: For /28 , $32 - 28 = 4$. Total IPs = $2^4 = 16$.
- **Usable Host Addresses:** Subtract 2 from the total (one for the Network ID and one for the Broadcast address).
 - Formula: $(2^{(32 - n)}) - 2$.
 - Example: For /24 , total IPs = 256; usable hosts = 254.
- **Subnet Mask Conversion:** The CIDR number represents the count of consecutive "1" bits in binary.
 - /8 = 255.0.0.0
 - /16 = 255.255.0.0
 - /24 = 255.255.255.0

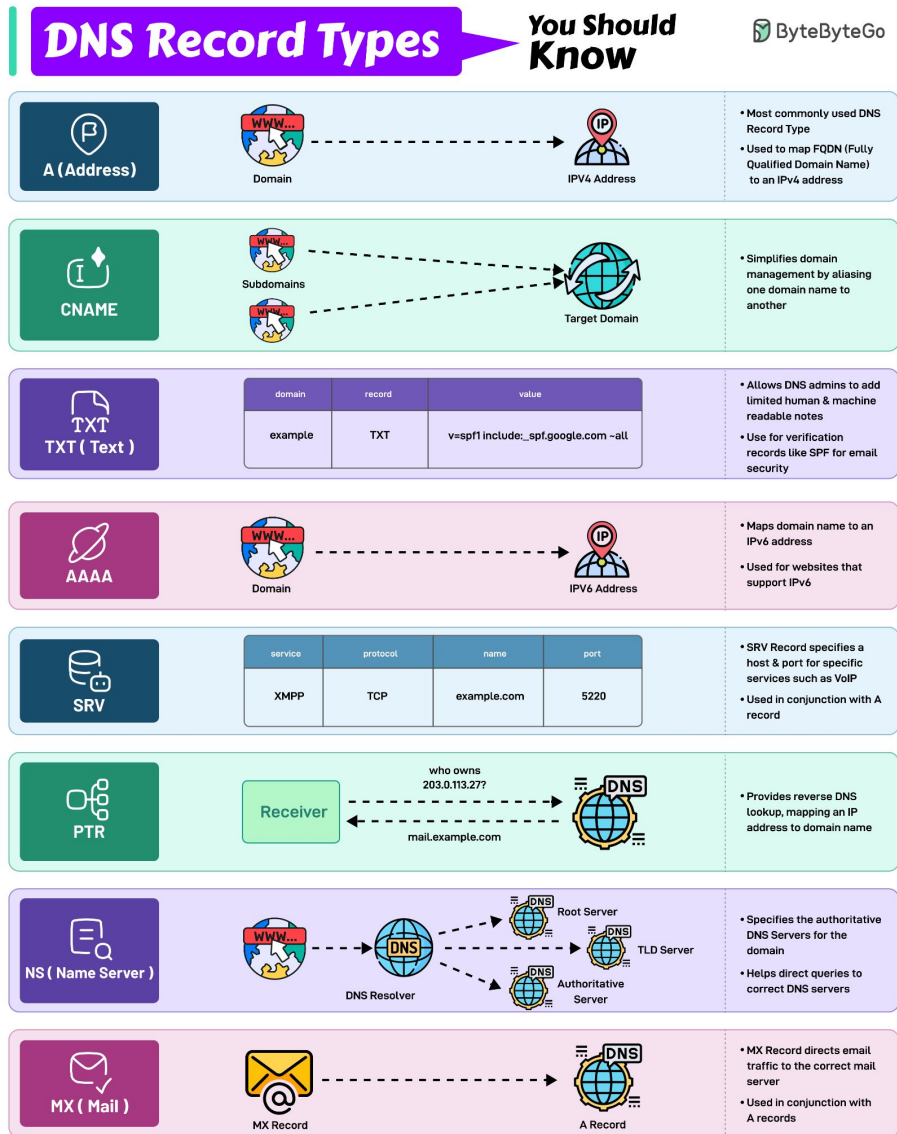
Common CIDR Blocks in AWS/GCP

- /16: The standard starting size for a Cloud VPC (65,536 IPs). Plenty of room to grow.
- /24: The standard size for an individual Subnet inside that VPC (256 IPs).
- /32: A single, specific IP address. Used in Security Groups to allow access from only one machine.

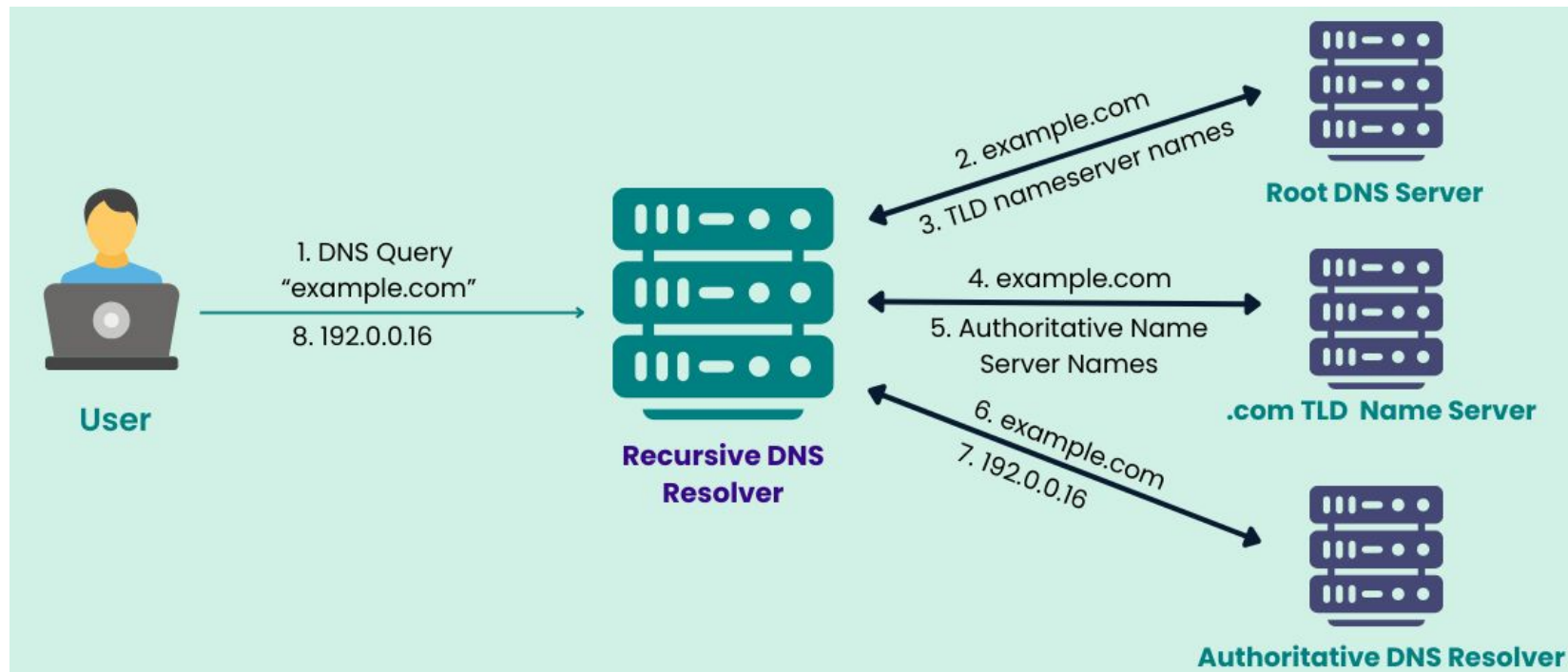


- Humans remember names (google.com); computers need numbers (142.250.190.46).
- Domain Name System (DNS) translates human-readable hostnames into machine-readable IP addresses.
- Without DNS, you would have to memorize the IP address of every API and database you use.



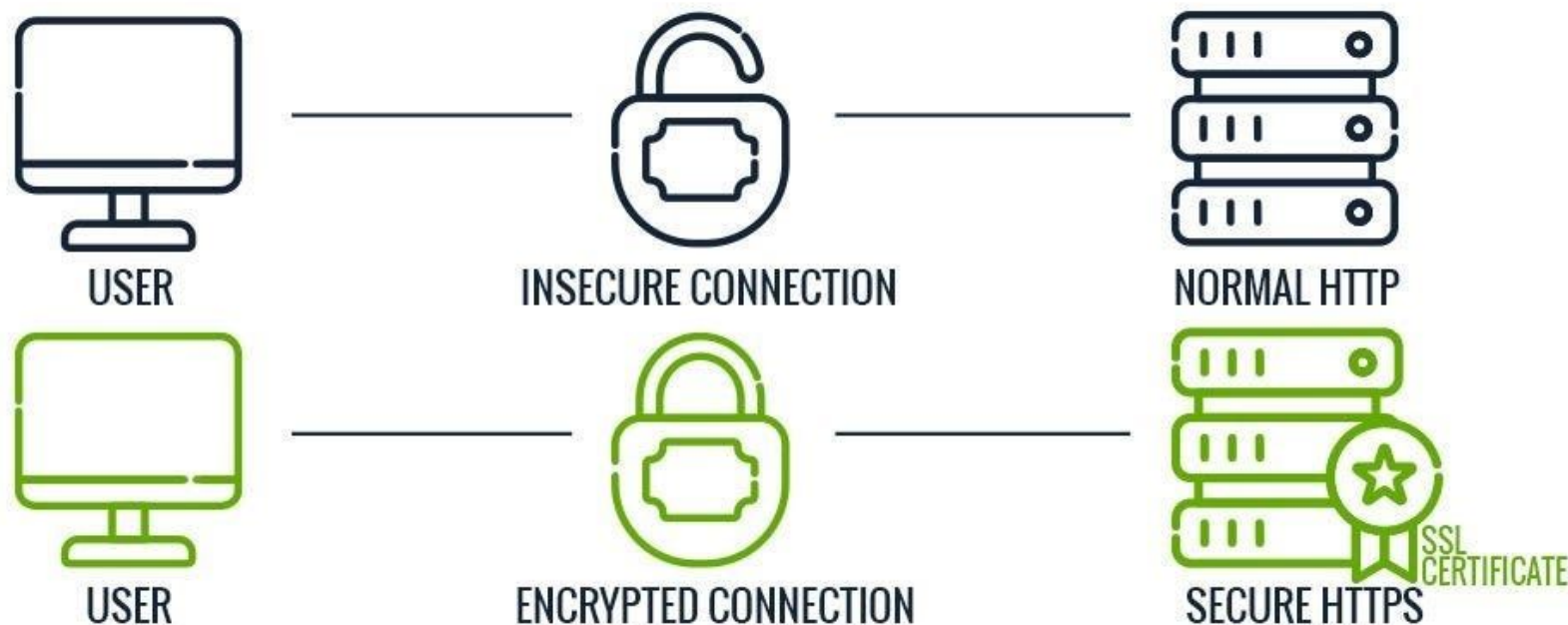


- Browser checks its local cache.
- OS checks the /etc/hosts file.
- Request goes to the Recursive Resolver (usually your ISP or Google's 8.8.8.8).
- Resolver traverses the Root Server -> TLD Server (.com) -> Authoritative Server to find the IP.

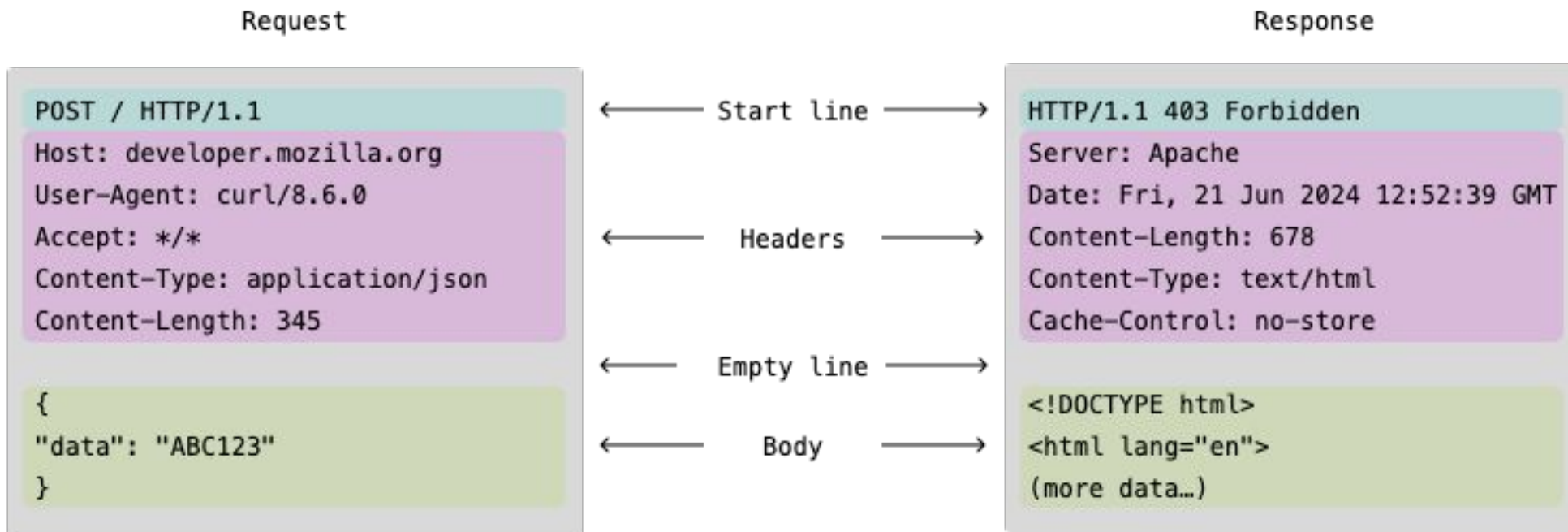


HTTP/HTTPS: The Language of the Web

- Hypertext Transfer Protocol is the foundation of data communication for the World Wide Web.
- HTTPS adds a layer of TLS/SSL encryption so data (like passwords) isn't sent in plaintext.
- When your Project Genesis app fetches data, it's making an HTTP request.



- Method: What action to perform (GET, POST, PUT, DELETE).
- URI: The specific path being requested (e.g., /api/v1/users).
- Headers: Metadata about the request (e.g., Authorization: Bearer <token>, Content-Type: application/json).
- Body: The actual data payload (used in POST/PUT).



- 2xx (Success): Everything went great (200 OK, 201 Created).
- 3xx (Redirection): Look somewhere else (301 Moved Permanently).
- 4xx (Client Error): You messed up (400 Bad Request, 401 Unauthorized, 404 Not Found).
- 5xx (Server Error): We messed up (500 Internal Server Error, 502 Bad Gateway).

1xx Informational	
100	Continue
101	Switching protocols
102	Processing
103	Early Hints

3xx Redirection	
301	Moved Permanently
302	Found
303	See Other
304	Not Modified

2xx Successful	
200	OK
201	Created
202	Accepted
203	Non-Authoritative
204	No Content
205	Reset Content
206	Partial Content
207	Multi-Status
208	Already Reported

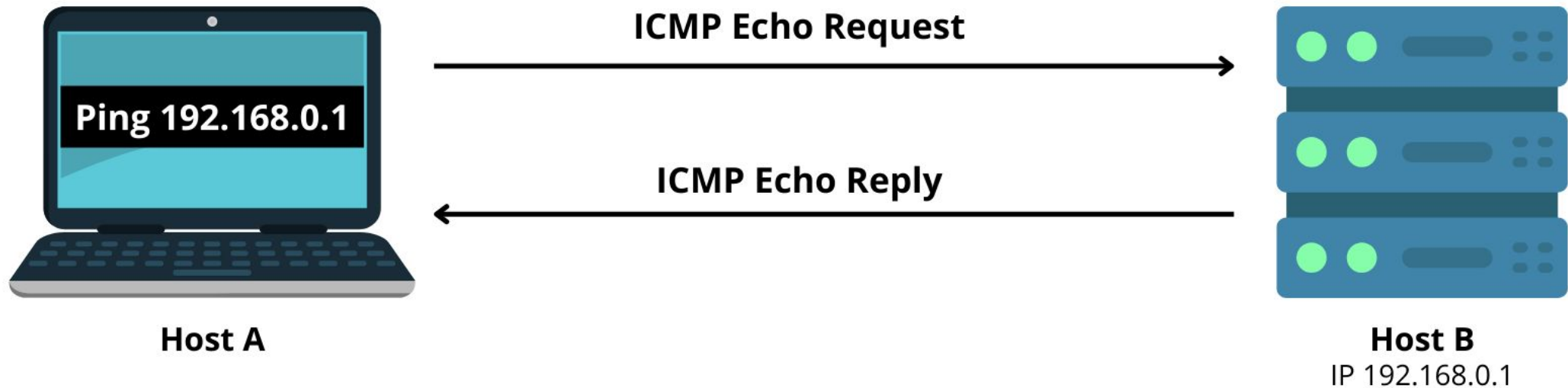
4xx Client Error	
400	Bad Request
401	Unauthorized
402	Payment required
403	Forbidden
404	Not Found
405	Method Not Allowed
406	Not Acceptable
409	Conflict
413	Payload Too Large
429	Too Many Requests

5xx Server Error	
500	Internal Server Error
501	Not Implemented
502	Bad Gateway
503	Service Unavailable
504	Gateway Timeout
505	HTTP Version Not Supported
506	Variant Also Negotiates
507	Insufficient Storage
508	Loop Detected
510	Not Extended
511	Network Authentication Required

ping & ip

- ping: Tests reachability to an IP using ICMP packets. (Note: Many cloud firewalls block ping by default!)
- ip a (Linux): Shows all network interfaces and assigned IP addresses on your machine.

```
ping 8.8.8.8  
ip a
```





- curl makes HTTP requests directly from the command line.
- The -v (verbose) flag is your best friend. It shows the complete DNS resolution, TLS handshake, and all HTTP headers sent and received.

```
marko@phoenixnap:~$ curl -d "name=test" https://httpbin.org/post
{
  "args": {},
  "data": "",
  "files": {},
  "form": {
    "name": "test"
  },
  "headers": {
    "Accept": "*/*",
    "Content-Length": "9",
    "Content-Type": "application/x-www-form-urlencoded",
    "Host": "httpbin.org",
    "User-Agent": "curl/8.5.0",
    "X-Amzn-Trace-Id": "Root=1-672a2dce-54cb99880dc7540d6cdce298"
  },
  "json": null,
  "origin": "188.2.111.125",
  "url": "https://httpbin.org/post"
}
```

- Used to query DNS name servers for information about host addresses.
- Crucial for verifying if a domain is pointing to the correct IP address or Load Balancer before tearing your hair out over application code.

```
khushi@ubuntu:~/geeksforgeeks$ dig geeksforgeeks.org ANY

; <<>> DiG 9.16.1-Ubuntu <<>> geeksforgeeks.org ANY
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 17954
;; flags: qr rd ra; QUERY: 1, ANSWER: 15, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;geeksforgeeks.org.          IN      ANY

;; ANSWER SECTION:
geeksforgeeks.org.          12      IN      A       34.218.62.116
geeksforgeeks.org.          17262   IN      NS      ns-1520.awsdns-62.org.
geeksforgeeks.org.          17262   IN      NS      ns-1569.awsdns-04.co.uk.
geeksforgeeks.org.          17262   IN      NS      ns-245.awsdns-30.com.
geeksforgeeks.org.          17262   IN      NS      ns-869.awsdns-44.net.
geeksforgeeks.org.          882     IN      SOA      ns-869.awsdns-44.net. awsdns-hostmaster.amazon.com. 1 7200 900 1209600 86400
geeksforgeeks.org.          282     IN      MX       1 aspmx.l.google.com.
geeksforgeeks.org.          282     IN      MX       10 alt3.aspmx.l.google.com.
geeksforgeeks.org.          282     IN      MX       10 alt4.aspmx.l.google.com.
geeksforgeeks.org.          282     IN      MX       5 alt1.aspmx.l.google.com.
geeksforgeeks.org.          282     IN      MX       5 alt2.aspmx.l.google.com.
geeksforgeeks.org.          282     IN      TXT      "fob1miabcp777bf2ncvnjn08n"
geeksforgeeks.org.          282     IN      TXT      "v=spf1 include:amazonses.com include:_spf.google.com -all"
geeksforgeeks.org.          282     IN      CAA      0 issue "godaddy.com"
geeksforgeeks.org.          282     IN      CAA      0 issue "letsencrypt.org"

;; Query time: 87 msec
;; SERVER: 127.0.0.53#53(127.0.0.53)
;; WHEN: Sun May 17 23:10:38 IST 2020
;; MSG SIZE rcvd: 548
```

```
student@Comp9:~$ nslookup -type=txt google.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
google.com       text = "v=spf1 include:_spf.google.com ~all"
google.com       text = "docusign=05958488-4752-4ef2-95eb-aa7ba8a3bd0e"
google.com       text = "facebook-domain-verification=22rm551cu4k0ab0bxs
w536tlds4h95"

Authoritative answers can be found from:
```

- Homework 8: Complete the previous assignments
- Next Lesson: DOCKER!

Questions?

CONTACT US